FPGAs4Quantum - ReadOut Experiment Documentation

Overview: QUANTUM READOUT

hauck@uw.edu , smouradi@uw.edu , geoffhjones@msn.com

Objective:

Threshold based readout of a Quantum experiment.
Refer thesis: m91941130511caa18e47dc606356dd66d

System:

Quantum Experiment → PMT → Analog Board → ARTIQ → Host-PC

Quantum Experiment - Dr. Sara's Lab, for us it is just a system we point our PMT to and expect a pulse every 100us or so….

PMT - TPMO1063E02
(So far, we have conducted PMT experiments with a make-shift blackbox, to observe pulses, the rate and width is okay for our ARTIQ to handle, refer meeting 02/13 for experiment data)

Analog Board - ADCMP600_601_602 - Analog Comparator - Google Docs, Hysteresis Setting for Comparators - Google Docs
Ethan and Pavan working on it, ref: pguntha@uw.edu
(Our PMT pulses pass through this to get converted into a square wave)

ARTIQ - Sinara core | M-Labs
We are using the UW-5 ARTIQ Box (Kasli 2.0, non SOC variant), Present in B023 Max Parsons Lab, need to request access to lab
(ARTIQ acts as the control center for our experiment we trigger from the host PC, runs on the soft CPU inside the Xilinx FPGA inside ARTIQ.
We will have to make gateware binary changes to ARTIQ's internal fpga to have a table for our threshold based counter, and depending on upper_count(timeSlot), lower_count(timeSlot), current_count, If we go beyond a threshold, stop experiment and save value into memory to be read later)

Logic to implement with gate-Ware changes (refer to thesis):
If value exceeds upper count → Quantum 0 & Stop Counting
If value exceeds lower count → Quantum 1 & Stop Counting
Else:
Keep counting

GateWare:
Written in migen, python library for SystemVerilog interface
OpenSource: M-Labs/artiq-zynq: ARTIQ Zynq-based core device support - artiq-zynq - M-Labs Git
(Current board is preFlashed with hardware too by the vendor)

More about ARTIQ:
Contains device_db.py which is a dataBase of the hardware resources, need to make a new one everytime we update the hardware. (Included the device_db.py for UW5 with the preFlashed gateware in the below drive)

Resources for scripting with ARTIQ (to install local libraries to start interfacing) - ARTIQ manual — ARTIQ 8.9024+a1bf0e1 documentation

(ARTIQ - whole system : SW+HW, Xilinx FPGA - inside ARTIQ box, Sinara Hardware - Manufacturer for ARTIQ hardware…)

Other resources (self, parallel projects at UW)
FPGAs4Quantum_Readout

Resources from previous student working on the same project :
FPGAs4Quantum Notes - Google Docs
(Note** :**Current project uses PMT instead of camera**)

Notes and Scripts from my work (refer to meetings for latest status of work):
FPGAs4Quantum

Students in Dr. Max Parsons Lab who might be of help: reynelc@uw.edu jul001@uw.edu